

NBA Player Points: Pre-Game Distributional Forecast

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1 What I built and why

I collapse the 5.49M possession-level rows to 59,256 player-game rows and produce a pre-game distributional forecast. The conditional mean is V3, a LightGBM regressor with Tweedie loss, trained on 40 numeric features. The variance layer is empirical: validation-fold residuals are binned by predicted mean and a per-bin NB2 dispersion parameter α is solved so that $\sigma^2 = \mu + \alpha\mu^2$. Over/under probabilities at any line L are then $1 - F(L)$ from the Negative Binomial CDF. The choice fell out of a head-to-head comparison (Section 3) against LightGBM with alternative losses, an Optuna-tuned variant, XGBoost benchmarks, a ZINB extension, and naive baselines.

2 Key decisions

Count distribution, not Gaussian. Player points are non-negative integers with a 12.3% zero mass and a heavy right tail (max 70). A Gaussian likelihood mis-specifies both the support and the shape. Tweedie loss (compound Poisson-Gamma, $p = 1.5$) places mass at zero, gives strictly non-negative predictions, and matches the skew. Tweedie marginally beats Poisson (MAE 4.380 vs. 4.396) and RMSE loss (4.419); loss choice is third-order but Tweedie is the right pick for this target distribution.

Strict leakage discipline. Every rolling and EWMA feature is computed within (entity, season) and shifted by one row so the current game never feeds its own feature. A unit test manually recomputes `pts_mean_season` and `pts_ewma_10` for random test rows and confirms they match. Train/val/test is strictly temporal (70/10/20 by date); adjacent player-games are highly autocorrelated, so random K-fold would leak future outcome information catastrophically.

40 features, all pre-game. Player form (EWMA of points, volume, usage, true-shooting%, minutes volatility, attempt share); context (home/away, rest days, back-to-back, three-in-four, position); team and opponent rolling 20-game stats (def-rating, off-rating, pace, shot volume allowed, points allowed to the player's 3-way position bucket); schedule (game index in season, month). Volume features are the strongest predictors, not efficiency.

Categoricals tested and rejected. Raw `team_id`, `opp_id`, and `position_code` as LightGBM categoricals are a no-op; the engineered rolling stats already absorb the team and position signal. Raw `player_id` actively hurts: MAE rises from 4.380 to 4.432 and CRPS from 3.068 to 3.099. The model memorises training-set players and fails to generalise across the 2024-25 test season where roles have shifted. Engineered features generalise better as they describe behaviour rather than identity.

No naive two-stage decomposition. An $\text{EWMA}(\text{minutes}) \times \text{EWMA}(\text{pts}/\text{min})$ baseline gives MAE 4.525, worse than plain EWMA-PPG (4.511). Multiplying two noisy estimates compounds error. A proper joint model would help most under regime shifts (a star resting, a backup promoted), but only with injury and lineup data this dataset does not contain. Minutes-related features (`min_ewma_10`, `min_std_10`, `starts_proxy_ewma_10`) are fed to the trees so LightGBM can learn the interaction directly.

3 Models compared

Variant (test, $n = 11,368$)	MAE	RMSE	CRPS
V1 10-game EWMA PPG (strong baseline)	4.511	5.984	n/a
V2 EWMA(min) $\times$ EWMA(pts/min) (naive 2-stage)	4.525	6.016	n/a
V3 LightGBM Tweedie (numeric only) [FINAL]	4.380	5.760	3.068
V4 LightGBM Poisson	4.396	5.764	3.070
V5 LightGBM RMSE	4.419	5.779	3.082
V6 LightGBM Tweedie + raw cats (pos/team/opp)	4.385	5.762	3.070
V7 LightGBM Tweedie + <code>player_id</code>	4.432	5.809	3.099
V8 LightGBM Tweedie (Optuna-tuned, 25 trials)	4.355	5.797	3.108
V9 XGBoost Tweedie	4.382	5.807	3.099
V10 XGBoost Poisson	4.402	5.761	3.070

Optuna tuning (V8). A 25-trial TPE study over 8 dimensions gives the best MAE (4.355, -0.6%) but worse CRPS (3.108 vs. 3.068). The tuned configuration (`num_leaves=201`, `min_data_in_leaf=54`) sharpens the conditional mean at the cost of variance calibration, the textbook bias-variance trade-off. For a sportsbook context where the full predictive distribution drives pricing, CRPS wins and V3 is kept. If the deliverable were point predictions only, V8 would win.

XGBoost (V9/V10). Both land in essentially the same neighbourhood as the corresponding LightGBM variants, confirming the result is not framework-specific. The predictive ceiling is feature-bound.

Zero-inflated NB tested, not adopted. The plain NB under-predicts the 12.3% zero mass (PIT residual spike at 0). A binary head is trained for $p_0 = P(Y = 0)$ and converted to a zero-inflation gate so the mixture's marginal $P(Y = 0)$ equals the trained p_0 . The empirical verdict: CRPS 3.074 (worse than plain NB's 3.068) and PIT KS-dev 0.064 (worse than 0.041). The gate shifts mass to 0 without preserving the marginal mean, which shrinks $P(Y > L)$

and over-corrects on most lines. A hurdle model (separate NB re-fit on $Y > 0$) is the unattempted version of zero-inflation, listed in future work.

4 Final headline

Model	MAE	RMSE	CRPS
Global-mean baseline	6.97	8.65	n/a
10-game EWMA PPG (strong baseline)	4.51	5.98	n/a
V3 LightGBM Tweedie + plain NB [FINAL]	4.38	5.76	3.068

Improvements vs. the EWMA baseline: MAE -2.9% , RMSE -3.7% . Over/under predicted probabilities are within ~ 1 pp of empirical at five of six lines tested; the exception is line 9.5 (-2.3 pp).

Player points are noisy at the game level. The irreducible floor comes from minutes variance, foul trouble, blowouts, and shooting variance, none of which are pre-game observable without external data. A 3% MAE reduction over EWMA is what an honest feature-engineering pass on this dataset can achieve; the head-to-head shows that bigger architectures (XGBoost), tuning (V8), and exotic distributions (ZINB) cannot materially move that floor.

5 Limitations

No DNP rows (model is conditional on appearance); no injury reports, lineup feeds, or market odds; opponent defence aggregated to 3-way position buckets only; mild calibration miss at line 9.5 (-2.3 pp); residual PIT spike at 0 (the ZINB fix did not help; hurdle model is the next attempt); player roles shift across seasons faster than EWMA can adapt.

6 What I would do with more time

- (1) **Hurdle model.** Separate NB re-fit on $Y > 0$, the unattempted version of zero-inflation that avoids the mean-preservation problem ZINB introduces. Most likely to close the PIT zero-spike.
- (2) **Temperature scaling.** A single Platt scalar trained across all lines to correct the systematic -2.3 pp under-prediction at line 9.5 and tighten calibration in the 0.3–0.6 predicted-probability range.
- (3) **Hierarchical shrinkage.** Shrink low-sample-size players (rookies, mid-season call-ups) toward their position and role priors to reduce high-variance point estimates on the test-season tail.
- (4) **MCMC-based hierarchical model.** Replace per-player EWMA with a full Bayesian posterior over player skill using MCMC (e.g. NUTS via NumPyro). This pools information across players with different sample sizes and produces principled uncertainty estimates rather than point-dispersion proxies.
- (5) **Quantile regression ensemble or conformal prediction intervals.** Non-parametric distributional alternatives that require no parametric family assumption, useful as a calibration cross-check against the NB layer.
- (6) **SHAP analysis.** Map per-feature contributions on the final LightGBM to verify that high-importance features behave sensibly across the full input range and to surface unexpected interactions.
- (7) **Possession-sequence player embeddings.** A small transformer over recent possession sequences to learn skill vectors that aggregate rate stats cannot capture, particularly for players whose value is concentrated in specific game situations.